Blockchain I
Cross-chain Tracking

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Ethereum is the largest public blockchain by usage and as such has become very expensive to use. By offering cheap transactions, many alternatives blockchains try to fill the gap left by Ethereum. Such a "multi-chain" world, has different implication on the user's privacy. We show to what extent it is possible to track the actions of users across different blockchain and build a user profile. We compose a dataset for Ethereum and another popular sidechain called Polygon and try to find relations between addresses. First, this is done in a general case to find general trends. Secondly, we focus on one address in particular and perform a case study. Additionally, we also study one particular application on both blockchains. While we were not able to find a strong indicator when users prefer what blockchain, we saw that increasing gas prices do not have a strong effect,

***Index Terms*—Blockchain, Cross-chain Tracking, Data Security, Smart Contracts, Interoperability, Privacy**

I. Introduction

Privacy in blockchains has been a much discussed topic since the inception of Bitcoin [1] and especially since the bust of the largest dark market place *Silk Road* [2]. In the past, many researcher focused on one blockchain in particular. This makes sense given that different blockchains have different privacy guarantees. Bitcoin utilizes an UTXO-based model to store user balances whereas Ethereum uses an account-based model, which comes with much weaker privacy guarantees [3].

In recent years, Ethereum has become more popular and is currently the largest blockchain by usage [4]. This is due to the large number of decentralized applications (dApps) that are being deployed and used on the Ethereum blockchain. dApps keep most of their application logic in smart contracts which are effectively programs on the blockchain. The two most popular categories of dApps are decentralized finance (DeFi), so financial application such as trading and taking on a loan, and non-fungible tokens (NFTs).

This has lead to a congestion of the Ethereum blockchain as the demand to issue transaction has outpaced the number of transaction the blockchain can process. As a result, the associated fees for issuing a transaction have increased significantly, which makes the blockchain unusable for many individuals as they are not able or willing to pay a high transaction fee.

There are many potential solutions to this problems. The Ethereum Foundation has been working on *Ethereum 2.0*, which will introduce sharding to the blockchain [5]. The hope is that sharding will help to drastically increase the scalability of the blockchain. Additionally, there are zk-rollups which utilize zero-knowledge proofs to compress transaction which results in a higher transaction throughput [6]. However, both solutions are still in development and therefore lacking adoption.

Therefore, a third solution has become popular. The source code of a popular Ethereum client, e.g. *Go Ethereum (Geth)*, is forked and adjusted to be more performant. This may include additional changes such as changing the consensus mechanism. This results in a new blockchain which offers higher scalability as Ethereum but as a trade-off is usually more centralized or less secure due to the blockchain trilemma [7].

Note that while the result is indeed a new blockchain, it fundamentally still works the same as Ethereum. That means it is able to run the same smart contracts, use the same address format and account-based model, and as such offers the same privacy guarantees. To be even more concrete, users can use the same wallet software to interact with these blockchains. To name some examples, Polygon, Avalanche and Fantom belong to this category of blockchains and are often called EVM-compatible blockchains, because they are all based on the Ethereum Virtual Machine (EVM), which is responsible for executing transactions among other things.

These separate blockchains are connected using a concept called bridges. Bridges allow users to transfer funds between different blockchains and in some cases even call smart contracts from a different blockchain [8]. This usually works by depositing some tokens on the origin blockchain and retrieving them on the target blockchain.

This opens up a new dimension for tracking users. It makes it possible to track the user's actions and associated funds across different blockchains. If the user's Ethereum address is know, we can easily check her funds on e.g. Fantom. This would not be possible for Bitcoin. Knowing the user's Ethereum address, reveals no information about her Bitcoin funds. As a result, users might not be aware how easy it is to connect their different "blockchain identities" if they use such compatible blockchains.

In this paper, we would like to shed light on the many ways users can be tracked between two blockchains, Ethereum and Polygon, and explore what we can learn about the user. In particular, we try to understand the user's behavior as well as preferences and build a rudimentary user profile. This is achieved by querying different blockchains and matching users based on their address but also by zooming in on particular applications that exist on both blockchains and analyze how users interact with them.

Note that this will not only give us insights about the user but also about the underlying blockchain. For instance, if we observe that most users move to Polygon to interact with DeFi applications, it might mean that this blockchain is particularly suited for DeFi as perceived by the users. So new DeFi applications should ideally be deployed on that blockchain first.

To summarize, our contributions are twofold

- We give a first glance of what's possible when it comes to cross-chain tracking.
- We show a simple way to build a user profile by taking into account different blockchains.

The paper is structured in the following way. After a literature review in section two, we explain our dataset and discuss how we queried the blockchain in section three. Section four will first elaborate on general insights we were able to gather by comparing the data between the two blockchains. Secondly, we pick on address in particular and analyze individual actions to build a user profile. We finish the section with an evaluating of our results. Finally we provide an outlook for future work and conclusion in section five.

II. Related Work

The increased gas prices opened up a huge room for research and development about improving the Ethereum blockchain and creating new blockchains. [9] is a crucial research about the factors causing the price increase. The surge in gas prices did not increase the user activity on every other platforms. For instance, [10] measures the statistical significance of increasing Ethereum gas price on blockchain-enabled Decentralized Autonomous Organizations (DAOs) and they suggest the effect is not statistically significant.

One of the first attempts to track users on the Bitcoin blockchain was done by [2]. They analyzed transactions related to *Silk Road*, an anonymous, international online marketplace that operates as a Tor hidden service and uses Bitcoin as its exchange currency.

In [11], two privacy preserving techniques are discussed that are added before the blockchain transaction data is published online. There are also various approaches to make interactions with smart contract more privacy preserving [12]–[14].

Often a user does not only have a single Ethereum address but multiple. This slightly increases privacy, because not everything can be learned about the user if only one address is known. However, [3] showed how graph representation learning, time-of-day activity and gas price based profiling can be used to link Ethereum addresses owned by the same user, thereby diminishing the privacy gains.

Bridges allow users to transfer funds across different blockchains but are usually pretty privacy invasive. In [15], a mechanism for privacy-preserving bridges is outlined by utilizing a mixer-like mechanism on the source and target blockchain.

III. Methods

A. Dataset Description

Blockchain networks are an alternative to centrally managed database systems and offer decentralized, distributed systems that runs on peer-to-peer networks. The term blockchain originates from the architecture of the network, which consists of chains of blocks. Each block in the sequence is connected with the previous block with a hash pointer. The blocks store information about transactions between individuals. However, these transactions are not limited to exchanging monetary values and can also allow an individual to execute an arbitrary code on the blockchain [16].

In order to access blockchain dataset in a convenient format, Ethereum-ETL [17], an open source public repository, was utilised to convert blockchain data into a relational databases. The Ethereum dataset consists of seven distinct tables which are interconnected to each user by foreign keys. The mentioned tables are: Blocks, Contracts, Logs, Token Transfers, Tokens, Traces, and Transactions. The Blocks table stores series of blocks with according timestamp. One block usually consists of multiple transactions. The Contracts table is used to store smart contracts which later are executed on the blockchain. The Logs table is used for reporting purposes, while the Token Transfers table stores transactions for ERC20 tokens. Basically, it is used to keep track of tokens that are moved from one address to another. The Tokens table stores information about each individual token that exists on the blockchain. The Trace table contains the information about the traces of program execution. Finally, the Transaction table stores transactions that happen from one address to another.

Cryptocurrency ledgers are usually very big in size. For instance, the research conducted on March 2021, concluded that the Bitcoin blockchain contained 674,001 blocks and 623,483,734 transactions [18]. According to Etherscan, the complete archive of blockchain dataset for Ethereum network consists of around 10TB worth of data. The size of the dataset poses various technical difficulties when working with blockchain data.

B. Dataset acquisition

In order to overcome the above-mentioned obstacle and query the huge dataset in reasonable time, we utilized Google BigQuery API. The Google BigQuery API allows users to run SQL queries on various Blockchain datasets. The subjects of interest for our project were Ethereum blockchain ledger and Polygon ledger, which are available through the BigQuery API. The BigQuery API is synchronized with mentioned blockchains and is updated daily. In addition, it is a multi-cloud data warehouse which partitions and stores data on multiple nodes in order to make it easily accessible.

C. Data Analytics

The numerical dataset which was acquired by running meaningful SQL queries on the Google BigQuery API, was analyzed using existing Python data analysis libraries (Pandas, Numpy, Matplotlib, Seaborn).

IV. Results

In this section we discuss bridging activities of users across different blockchains, then we analyze the Aave protocol, and finally we propose a case study in which a specific user address is selected and all the transactions are analysed. This section consists of the following experiments:

1) Both Ethereum and Polygon are analysed, in order to find a correlation between transaction cost increase on Ethereum and an increase in activity (total amount traded) on Polygon.
2) The types of transactions users perform on the Aave protocol is analysed and the borrowing trend for December 2020 is investigated.
3) A case study is performed in which the activity of a one randomly selected individual is analysed.

A. Bridging Activity

One of questions that we intend to answer with our research is to figure out the bridging activities of the individuals. We assume that when the transaction cost on Ethereum Mainnet increases, individuals will use sidechains such as Polygon to perform their transactions on instead.

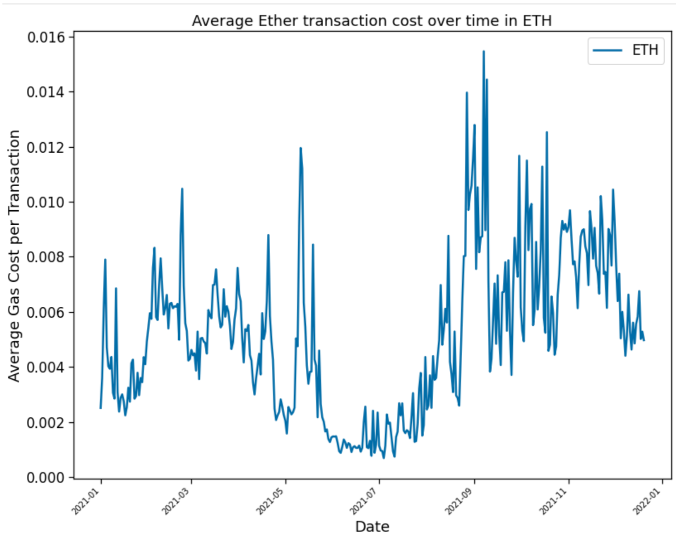

Fig. 1: Average transaction cost over time for Ethereum measured in Ether.

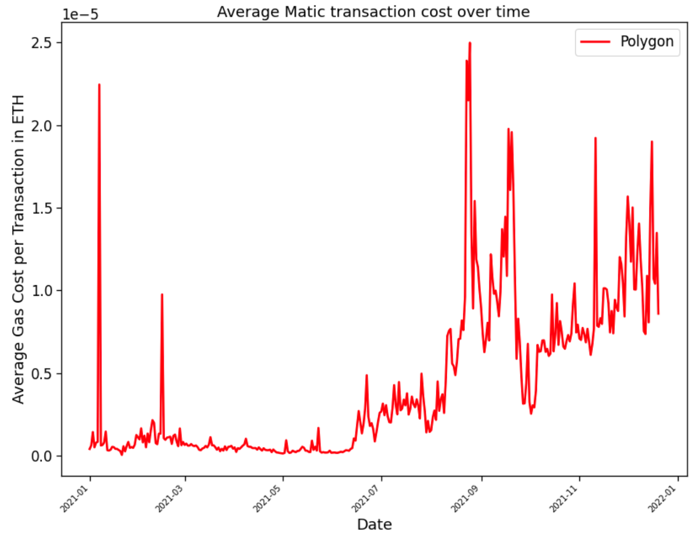

Fig. 2: Average transaction cost over time for Polygon measured in Ether.

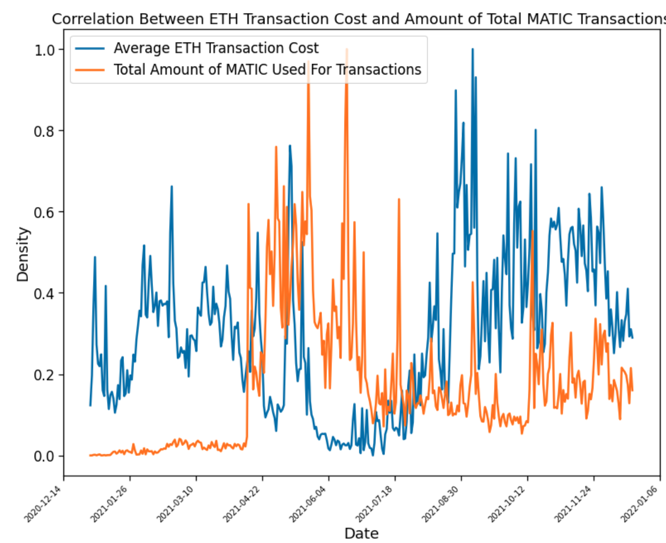

Fig. 3: The comparison between average transaction cost on Ethereum and the total amount of trades conducted on Polygon.

In order to capture relationships between the Ethereum and Polygon blockchain, we performed quantitative analysis on the outputs of Google BigQuery API. Figure 1 shows the average transaction cost over time for Ethereum, while Figure 2 demonstrates similar metrics for Polygon. The trades on Polygon are performed in Matic, which is a native currency for that blockchain. However, to make figures comparable, the mentioned currency was converted to Ether. Consequently, the y-axis for the both of the figures represent transaction costs in Ether (ETH). Furthermore, it can be observed in these figures, that the average costs on Ethereum are significantly more in value than those on Polygon. Thus, it could lead people to perform trades on Polygon.

In order to validate our above-mentioned assumption, we expect that the increase in transaction cost on the Ethereum blockchain would also increase the total amount of ETH used on Polygon. The Figure 3 portrays the correlation between these metrics. In addition, the Figure is color coded and average ETH transaction cost is represented in blue, while orange is denoted for the total amount of Matic that is used for transactions. We expected a positive correlation between these metrics, meaning that the increase in ETH transaction cost would result in an increase in the total amount traded on the Polygon sidechain. However, even though this assumption seems sound, we found out that these two measures are not strongly correlated, and in order to reach a reasonable conclusion we would need to explore user behaviour on other sidechains too.

B. Aave Protocol

Aave is a decentralized liquidity market protocol where users can participate as depositors or borrowers. The Aave protocol is using a pool-based strategy, which differentiates itself from other P2P lending protocols.

In this section, we analyzed user behaviour on the Aave protocol. In this case study, we point out types of transactions that are performed on this protocol and we intend to showcase borrowing trends over time. The dataset for Aave protocol is not available on Google BigQuery platform, thus Aave's native website was utilized to acquire the dataset, however the dataset is not extensive and it does not contain information for recent times. Even though, we could not acquire most recent data for this protocol, we still deem it worthwhile to analyze the data for December 2020, because it can give us insights about how this protocol was used in the early stages of development.

We analyzed the transactions on Aave protocol that happened in the December of 2020. There are 12 different types of transactions that users can perform on this protocol. Figure 4 shows types of transactions on x-axis and relative amount in ETH on y-axis. The most frequent transaction was money borrowing, followed with money deposit and in total people borrowed around 30 ETH worth of money in December 2020.

In addition, we wanted to investigate the borrowing trends for this protocol. Nowadays, the total market size of Aave protocol is worth around 10 million US Dollars, however as we can see from figure 4 the cash flow on this protocol was nowhere near the mentioned amount, so we assume that over time people borrowed more money.

Figure 5 showcases the borrowing trend over time for December 2020. The y-axis represents the total amount that was borrowed, while x-axis is denoted for dates. We could clearly observe a slight upward trend at the end of the month, however it is hard to deduce what was the cause of the upward movement. It is hard to tell whether the increase happened due to the increased number of new users or it happened because it was Christmas time.

At this point, we only had access to the one-month worth of data, consequently in order to reach sound conclusion, it is required to analyze data for other dates.

C. Case Study

To understand the users' behavior between different blockchains, transactions of a specific user are investigated. All transfers of tokens done in 2021 are fetched by using Google BigQuery. Then, users are ranked by their number of transactions and a random user who has many transactions is picked. Note that we only know the address of the user and her financial activity, nothing about her name or personal details.

All the token transfers of that user were queried and the tokens that she mostly bought and sold were investigated. Tether USD is the token that she bought and sold most, second is the ChainLink token and third is Molecular Future token. Then, we analyzed which tokens she bought more than she sold to see what she still holds. Molecular Future, KOK Coin and Universal Euro are one of them. Figure 3 shows the network diagram of ten tokens that she bought most. If the arrow is from token to "Account" it means she bought the token more than she sold, so the token still exists in her account and it is vice versa if the arrow is from account to token.

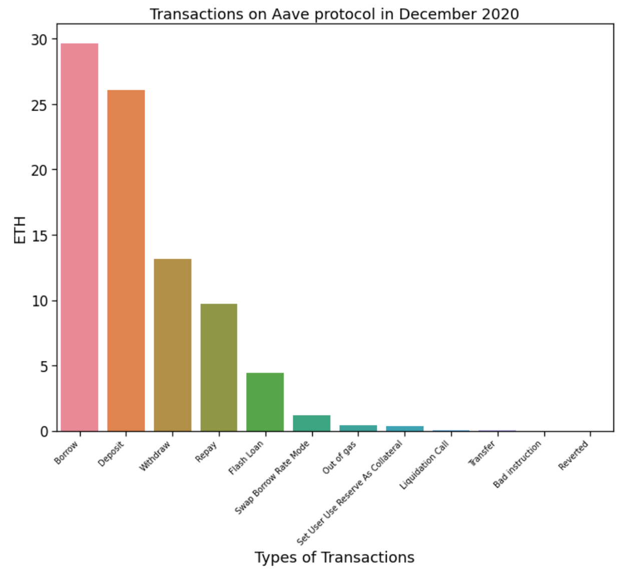

Fig. 4: Types of unique transactions that are conducted using Aave protocol.

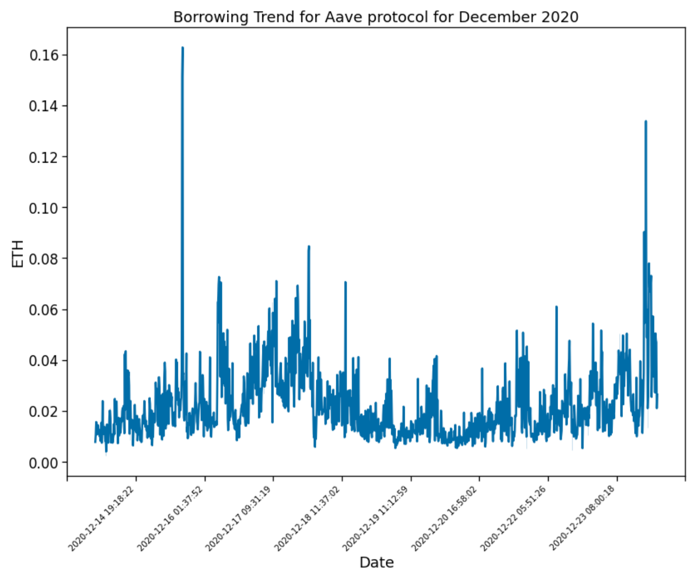

Fig. 5: The total amount borrowed over time using Aave protocol.

Due to not having many transactions in the token Universal Euro (UPEUR), we decided to analyze her behavior on this token. The figures below show the user's transactions and the price chart of Universal Euro. She sometimes buys and sells on the same day and this occurs in some other tokens very often. This shows that the investments are short-term and the purpose is mostly trading instead of investing. As it is seen from the table, she bought Universal Euro in the first quarter of the year multiple times. Then, she started to sell slowly, but on August 22, when she sold a big amount of token, it was actually at one of its lowest values. The maximum amount of token was sold on September 2 and there was a 60 percent increase in the value on that day. However, if she would have waited one more week, the increase would have been 165 percent.

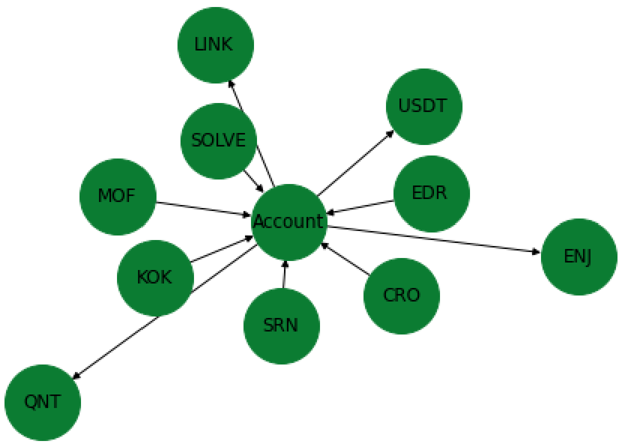

Fig. 6: The tokens that selected account traded with

Therefore, the behavior on these transactions are not always understandable since some behaviors are not logical in terms of making a financial profit. As it is seen in the whole analysis, the blockchain network there is a lot of valuable data that can be analyzed for an account, such as previous transactions and the account's Ether balance. Nevertheless, since the account owner is unknown, the person's risk-aversion, the need of money at a specific time and many other factors are unknown by the analyzer. Because of these unknown factors, results and interpretations about financial behavior of a specific user may not always be certain.

The address of the selected account is also queried in Polygon network. We found that she does not sell any tokens on Polygon in 2021 and she only bought CryptoMETH seven times. Moreover, we looked for ten different addresses which had the most token transfers on Ethereum and none of them had more than ten token transfers on the Polygon network. In addition, there is no selling token transaction in nine of these addresses at all. Only tokens that they bought are CrpytoMETH and Wrapped Matic. Hence, we can conclude that the accounts who do the most token transfers on Ethereum do not do token transfers often on Polygon, despite the increase in gas prices.

date	value
2021-01-02 19:41:04+00:00	-99700
2021-01-05 20:28:05+00:00	50000
2021-01-11 23:51:44+00:00	19573
2021-01-16 21:03:28+00:00	-17473
2021-01-21 18:47:04+00:00	-137546
2021-01-29 22:40:54+00:00	77157
2021-01-31 13:01:32+00:00	443
2021-03-09 07:04:18+00:00	219211
2021-03-22 22:31:54+00:00	242960
2021-04-01 12:38:13+00:00	413486
2021-04-01 13:21:51+00:00	413141
2021-05-18 18:22:13+00:00	-296800
2021-08-22 07:16:39+00:00	-298700
2021-10-01 22:47:30+00:00	-7200
2021-10-02 08:49:53+00:00	-854078

(a) Transactions on Universal Euro of the selected account

(b) Universal Euro Price Chart

Fig. 7

D. Evaluation

Our implementation details are evaluated in the following section regarding the time and cost for each query. First of all, the free version of Google BigQuery API is limited in the memory that can be accessed each days. For instance, it is only possible to process 50 terabytes per day and due to the fact that blockchain dataset in huge in size, we were required to use trial version of premium membership. Furthermore, for our purposes the proposed trial processing plan was sufficient. Each query on average was processing around 300GB worth of data and the execution time on average was 2 minutes. The implementation details is publicly available via the following Github repository [19].

In the section IV-A, we found out that the connection between average transaction cost on Ethereum and the increased activity on Polygon, is not trivial. No direct positive correlation was observed between these two metrics and in order to understand the bigger picture of user behaviour, we think it is worthwhile to include other side-chains into our analyzes. As it was mentioned above, the Ethereum Network interacts with dozen of other side-chains, thus there are many more variables that need to be considered in order to evaluate our assumption.

There is a lot of data recorded about a user's activity in blockchain networks. The user's account balance and many details about these transactions such as the time of the transaction, the address they bought from and the block number that the transaction took place in. In the case study part, we successfully retrieved all the transactions of a selected user and tracked the account's transactions. By analyzing these data, we were able to understand some financial behaviors of the user such as how frequent she trades, how much balance she has, which tokens she trades with, which tokens she holds for long term.

However, it is not possible to conduct a characteristic analysis for the user due to the anonymity of the personal data. We fundamentally know only the address of the user, no personal information could be gathered directly. Thus, findings about patience or risk-aversion of the user is impossible, although some analysis showed that the user could increase earnings by only waiting a week.

We couldn't perform advanced statistical analysis between Ethereum and Polygon for the selected user since the user basically did not have many transactions on Polygon. However, this also answers the question that we were initially looking for. Our analysis shows that the users who do most token transfers in Ethereum did not move to Polygon, despite the surge in gas prices.

V. Conclusion

In this paper, we have queried the Ethereum and Polygon blockchain and analyzed how we can link data points from either blockchain to a singe address. Additionally, we have look at general trends regarding both blockchains.

We were not able to find a strong indicator when users prefer to use what blockchain. Furthermore, we saw that the trading behavior of a given address can be difficult to interpret and additional analysis might be needed.

In the future, it would be interesting to take a closer look at projects such as Tornado cash, which have to goal to increase privacy [20]. They might not only increase privacy on a single blockchain but also across multiple blockchains.

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